Response to Reviver

Title: ImageVault: A Decentralized Blockchain-based Image Management Platform
Paper ID: ICAIS-80
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We are thankful to all the Reviewers for spending their precious time and providing positive and constructive comments on our manuscript that leads to further improvements in the quality of the paper. We have made our best effort to address all the concerns raised by the Reviewers.

Review Comments

Review Comments 1:

1. Revise the abstract to clearly distinguish the proposed ImageVault architecture from existing systems.

We appreciate your concern, regarding this, we have revise the abstract. The revised abstract is:

In today's digital age, most existing image storage systems are built on centralized cloud infrastructure or partially decentralized systems. These systems expose sensitive personal and corporate images to risks such as privacy (unauthorized access, single point failure) and integrity (data tempering). While blockchain technology primarily mitigates these risks, it focuses on hash-based security and immutability. To inspire with this technology, this work proposes a decentralized and distributed image storage platform to secure images and preserve their privacy and integrity. It uses the InterPlanetary File System and blockchain technology. To protect ownership and access privileges of images, the platform integrates IPFS's content-addressed storage and blockchain's immutability. This Ethereum-based solution ensures seamless owner approvals, access requests, and image uploads without intermediaries. To offer strong security and trust, standout features include transparent access restriction, local encryption, and immutable transaction recording. This research shows how blockchain technology can be integrated with IPFS to enhance user trust in managing sensitive digital content, protect private photographs, and eliminate the need for centralized storage. The proposed system addresses the limitations of centralized and semi-decentralized image storage systems. This proposal is capable to secure all types of images, which means it is not limited to images from a specific domain.

2. Please clarify the problem statement, list out the concrete security and access-control gaps addressed by ImageVault.

Thank you for your suggestion. As per suggestion, we have incorporated a paragraph at the end of the Introduction Section. The added lines are:

The proposed system addresses the security and privacy limitations of centralized systems, including a single point of failure, lack of transparent access governance, and reliance on trusted third parties to enforce ownership and access policies. It also improves the visibility of image owners (i.e., how their images are accessed) and their control over access revocation. It provides an immutable audit trail of image usage to owners. In short, ImageVault provides a decentralized, owner-controlled image storage platform that combines encrypted off-chain storage with blockchain-enforced access governance, delivering privacy preservation, tamper resistance, and verifiable accountability.

3. In the system architecture, please mention which components operate on-chain versus off-chain

Thank you for your suggestion. As per suggestion, we have incorporated few lines in the second last paragraph of the Proposed System Architecture Section. The updated paragraph is:

This on-chain and off-chain storage mechanism provides a scalable, cost-efficient framework for users. In this framework, BCNw (i.e., on-chain) stores only metadata and access logs of images, while the images themselves are stored in IPFS (i.e., off-chain) in encrypted format~\cite{patil2024secure}. It reduces gas costs and supports large-scale deployment. Furthermore, the modular framework can extend beyond the image storage to other sensitive digital assets. These digital assets can be educational certificates, legal documents, property records, supply chain records, and medical records. This hybrid on-chain/off-chain concept is designed to balance security, transparency, and scalability. Computationally expansive and large data operations are handled off-chain, while critical access-control and audit functions are enforced on-chain. The architecture of the proposed system is illustrated in Figure 1.

4. Improve Figure 1 by adding clearer labels for encryption, key exchange and audit logging steps in the workflow.

We appreciate the reviewer’s valuable concern. However, at the current stage of this work, we are unable to incorporate further improvements. We will enhance the figure with more fine-grained details and comprehensive information in our subsequent, extended version of the work.

5. Revise the results section to include comparative performance or cost metrics against at least one existing system.

We thank the reviewer for highlighting this important point. Due to the current scope of the work and associated constraints, a quantitative performance or cost evaluation against existing systems could not be incorporated at this stage. We regret this limitation and have clearly identified it as a future research direction in the Future Directions section.

6. Result images should be in a result section, kindly check and update the result section appropriately. It is recommended that the conclusion and future scope be presented as a single paragraph. Kindly revise the manuscript to reflect this requirement.

Thank you for your suggestion. Necessary corrections have made in conclusion and future direction section. The result images are aligned in result section. The updated conclusion and future direction section is:

This paper presents the concept, functionality, and initial evaluation of ImageVault, a decentralized platform for secure image storage. This platform utilizes blockchain for controlled sharing, IPFS for distributed file storage, and client-side encryption for data privacy. The system encrypts the images locally before uploading them to IPFS, and stores only the metadata and access policies on the Ethereum blockchain. In the system, the smart contract manages the ownership, access control, and transaction log. The public-key cryptography technique enables secure key exchanges between users. The Angular-based front end, integrated with web3.js or ethers.js, provides an intuitive interface for managing uploads, approvals, and access requests. This proposed system demonstrates the practical integration of decentralized technologies to develop privacy-preserving, transparent, and scalable image management systems which is suitable for various real-world applications. ImageVault offers a decentralized, owner-centric image storage platform that integrates encrypted off-chain storage with blockchain-enforced access control, ensuring privacy preservation, tamper resistance, and verifiable accountability. This proposed system requires a more in-depth evaluation in a real-world scenario and requires comparison with existing systems. A formal analysis of the results is also required to demonstrate the security and robustness of the system. The current work lacks a detailed discussion of the threat model and potential weaknesses of the system.

Review Comments 2:

1. Clarify the main contribution of the paper in the introduction and conclusion.

We appreciate your concern, regarding this, we have improved the introduction and conclusion section. These improvements are:

Introduction:

In today's digital age, most existing image storage systems are built on centralized cloud infrastructure or partially decentralized systems. These systems expose sensitive personal and corporate images to risks such as privacy (unauthorized access, single point failure) and integrity (data tempering). While blockchain technology primarily mitigates these risks, it focuses on hash-based security and immutability. To inspire with this technology, this work proposes a decentralized and distributed image storage platform to secure images and preserve their privacy and integrity. It uses the InterPlanetary File System and blockchain technology. To protect ownership and access privileges of images, the platform integrates IPFS's content-addressed storage and blockchain's immutability. This Ethereum-based solution ensures seamless owner approvals, access requests, and image uploads without intermediaries. To offer strong security and trust, standout features include transparent access restriction, local encryption, and immutable transaction recording. This research shows how blockchain technology can be integrated with IPFS to enhance user trust in managing sensitive digital content, protect private photographs, and eliminate the need for centralized storage. The proposed system addresses the limitations of centralized and semi-decentralized image storage systems. This proposal is capable to secure all types of images, which means it is not limited to images from a specific domain.

Conclusion:

This paper presents the concept, functionality, and initial evaluation of ImageVault, a decentralized platform for secure image storage. This platform utilizes blockchain for controlled sharing, IPFS for distributed file storage, and client-side encryption for data privacy. The system encrypts the images locally before uploading them to IPFS, and stores only the metadata and access policies on the Ethereum blockchain. In the system, the smart contract manages the ownership, access control, and transaction log. The public-key cryptography technique enables secure key exchanges between users. The Angular-based front end, integrated with web3.js or ethers.js, provides an intuitive interface for managing uploads, approvals, and access requests. This proposed system demonstrates the practical integration of decentralized technologies to develop privacy-preserving, transparent, and scalable image management systems which is suitable for various real-world applications. ImageVault offers a decentralized, owner-centric image storage platform that integrates encrypted off-chain storage with blockchain-enforced access control, ensuring privacy preservation, tamper resistance, and verifiable accountability. This proposed system requires a more in-depth evaluation in a real-world scenario and requires comparison with existing systems. A formal analysis of the results is also required to demonstrate the security and robustness of the system. The current work lacks a detailed discussion of the threat model and potential weaknesses of the system.

2. Include a more comprehensive related work section and contrast your approach with others.

We appreciate on raised concern, but Due to the current scope of the work and associated constraints, we are unable to enhance the related work section. We really apologize on it.

3. Suggest to include some relevant and recent references.

Thank you for your suggestion. As per suggestion, we have incorporated some latest references.

4. The novelty of the proposed approach is to be demonstrated in detail.

As per the suggestion, we have improved the abstract, introduction, proposed system architecture and conclusion section with the novelty prespective.

5. Include a clear justification for the selection of the proposed approach.

As per the suggestion, we have improved the abstract, introduction, proposed system architecture and conclusion section.

6. Revise the figures/tables for clarity and resolution. Ensure all the figures and tables have correct caption and cited in the text accordingly.

Checked and necessary corrections have made.